

## Advanced challenge

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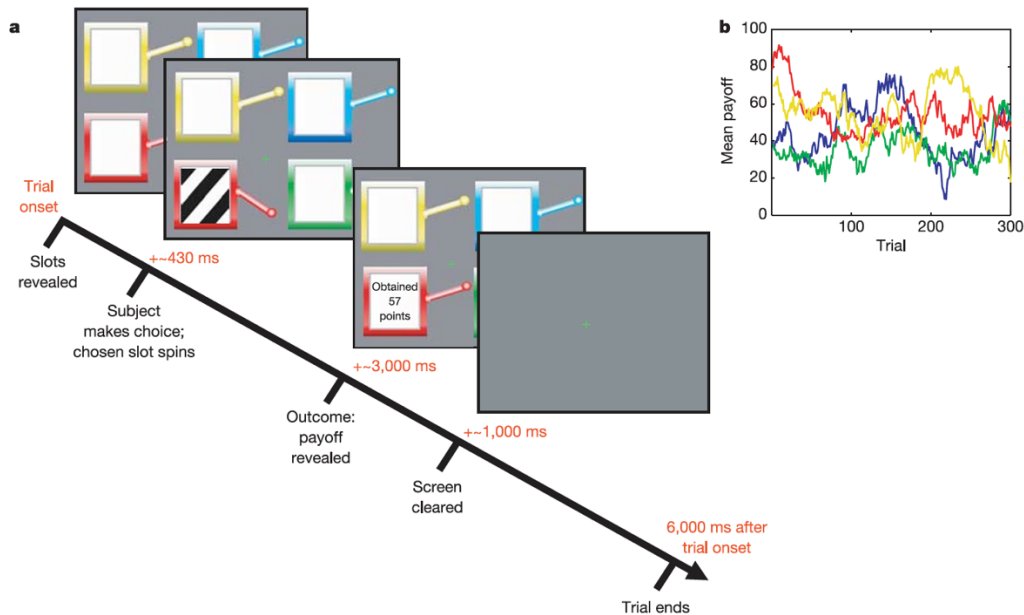
1. Initialize a stationary 4-armed bandit task environment (use “NArmBandit” class) to return binary rewards and then perform simulations with the following agents:

- (a) Agent estimating a value by using a fixed learning rate
- (b) Agent estimating a value distribution using Bayesian methods (hint: Bernoulli-Binomial conjugacy, a.k.a. Thompson sampling)

For agents (a), you can use an adaptive epsilon-greedy action policy with three arguments: initial epsilon, decaying factor of the epsilon per trial, and minimum epsilon. For each of the two agents:

- (1) Show how arms' values are learned. For agent (b), you also need to show the variance of the learned arms' values.
- (2) Compare the learning curves of the two agents and discuss the results.

2. Initialize a 4-armed bandit task environment (use “NArmBandit” class) to return continuous and nonstationary rewards and then perform simulations with the following agents:



Daw et al., *Nature* 2006

(1) Implement an agent using the Kalman filter to estimate a non-stationary value function of each arm for 500 trials. Assume that the agent knows the variance of the non-stationary drift of the mean payoffs from 4 arms (“state\_noise” of “NArmBandit” class) and the variance of the sampled rewards from the 4 arms (“measurement\_noise” of “NArmBandit” class). You can use any action policy (e.g., epsilon-greedy, softmax, and so on) for the agent.

(2) Plot time-varying estimated mean rewards with an overlay of the actual mean rewards. For the plots, you should show the estimated posterior variance from the Kalman filter using shaded error bars (estimated  $mean \pm \sqrt{posterior\ variance}$ ).

(3) Repeat (2) with different values of “state\_noise” “measurement\_noise” and discuss the results.

## Reference

- Daw, N. D., Niv, Y., & Dayan, P. (2005). Uncertainty-based competition between prefrontal and dorsolateral striatal systems for behavioral control. *Nature neuroscience*, 8(12), 1704-1711.
- Daw, N. D., O'Doherty, J. P., Dayan, P., Seymour, B., & Dolan, R. J. (2006). Cortical substrates for exploratory decisions in humans. *Nature*, 441(7095), 876-879.